MSG No. 227.130 $Fd'\bar{3}'m$ [Type III, cubic]

Table 1: Wyckoff site: `8a`, site symmetry: `-43m`

No.	position	mapping
1	$\left[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]
2	$\left[\frac{3}{8}, \frac{7}{8}, \frac{3}{8}\right]$	[25,30,35,39,43,48,74,75,79,88,92,93,128,130,132,133,137,138,172,173,177,182,190,191]
3	$\left[\frac{3}{8}, \frac{3}{8}, \frac{7}{8}\right]$	[26,27,31,40,44,45,73,78,83,87,91,96,124,125,129,134,142,143,176,178,180,181,185,186]
4	$\left[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}\right]$	[28,29,33,38,46,47,80,82,84,85,89,90,122,123,127,136,140,141,169,174,179,183,187,192]
5	$\left[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}\right]$	[32,34,36,37,41,42,76,77,81,86,94,95,121,126,131,135,139,144,170,171,175,184,188,189]
6	$\left[\frac{1}{8}, \frac{5}{8}, \frac{5}{8}\right]$	[49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72]
7	$\left[\frac{5}{8}, \frac{1}{8}, \frac{5}{8}\right]$	[97,98,99,100,101,102,103,104,105,106,107,108,109,110,111,112,113,114,115,116,117,118,119,120]
8	$\left[\frac{5}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[145,146,147,148,149,150,151,152,153,154,155,156,157,158,159,160,161,162,163,164,165,166,167,168]

Table 2: Wyckoff site: `8b`, site symmetry: `-43m`

No.	position	mapping
1	$\left[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}\right]$	[1,5,6,20,22,24,50,58,59,64,65,69,99,103,108,109,114,119,148,152,153,158,159,163]
2	$\left[\frac{3}{8}, \frac{7}{8}, \frac{7}{8}\right]$	[2,10,11,16,17,21,49,53,54,68,70,72,100,104,105,110,111,115,147,151,156,157,162,167]
3	$\left[\frac{7}{8}, \frac{3}{8}, \frac{7}{8}\right]$	[3,7,12,13,18,23,52,56,57,62,63,67,97,101,102,116,118,120,146,154,155,160,161,165]
4	$\left[\frac{7}{8}, \frac{7}{8}, \frac{3}{8}\right]$	[4,8,9,14,15,19,51,55,60,61,66,71,98,106,107,112,113,117,145,149,150,164,166,168]
5	$\left[\frac{5}{8}, \frac{5}{8}, \frac{5}{8}\right]$	[25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]
6	$\left[\frac{5}{8}, \frac{1}{8}, \frac{1}{8}\right]$	[73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96]
7	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[121,122,123,124,125,126,127,128,129,130,131,132,133,134,135,136,137,138,139,140,141,142,143,144]
8	$\left[\frac{1}{8}, \frac{1}{8}, \frac{5}{8}\right]$	[169,170,171,172,173,174,175,176,177,178,179,180,181,182,183,184,185,186,187,188,189,190,191,192]

Table 3: Wyckoff site: `16c`, site symmetry: `.-3'm`

No.	position	mapping
1	$[0, 0, 0]$	[1,5,6,20,22,24,32,34,36,37,41,42]
2	$\left[0, \frac{1}{4}, \frac{1}{4}\right]$	[2,10,11,16,17,21,28,29,33,38,46,47]
3	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	[3,7,12,13,18,23,25,30,35,39,43,48]
4	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	[4,8,9,14,15,19,26,27,31,40,44,45]
5	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[49,53,54,68,70,72,80,82,84,85,89,90]
6	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	[50,58,59,64,65,69,76,77,81,86,94,95]
7	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[51,55,60,61,66,71,73,78,83,87,91,96]
8	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[52,56,57,62,63,67,74,75,79,88,92,93]
9	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[97,101,102,116,118,120,128,130,132,133,137,138]
10	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[98,106,107,112,113,117,124,125,129,134,142,143]
11	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	[99,103,108,109,114,119,121,126,131,135,139,144]
12	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[100,104,105,110,111,115,122,123,127,136,140,141]
13	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[145,149,150,164,166,168,176,178,180,181,185,186]
14	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[146,154,155,160,161,165,172,173,177,182,190,191]
15	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[147,151,156,157,162,167,169,174,179,183,187,192]
16	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	[148,152,153,158,159,163,170,171,175,184,188,189]

Table 4: Wyckoff site: `16d`, site symmetry: `.-3'm`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[1,5,6,20,22,24,32,34,36,37,41,42]
2	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[2,10,11,16,17,21,28,29,33,38,46,47]
3	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[3,7,12,13,18,23,25,30,35,39,43,48]
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[4,8,9,14,15,19,26,27,31,40,44,45]
5	$\left[\frac{1}{2}, 0, 0\right]$	[49,53,54,68,70,72,80,82,84,85,89,90]
6	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	[50,58,59,64,65,69,76,77,81,86,94,95]
7	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	[51,55,60,61,66,71,73,78,83,87,91,96]
8	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[52,56,57,62,63,67,74,75,79,88,92,93]
9	$\left[0, \frac{1}{2}, 0\right]$	[97,101,102,116,118,120,128,130,132,133,137,138]
10	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	[98,106,107,112,113,117,124,125,129,134,142,143]
11	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[99,103,108,109,114,119,121,126,131,135,139,144]
12	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[100,104,105,110,111,115,122,123,127,136,140,141]
13	$\left[0, 0, \frac{1}{2}\right]$	[145,149,150,164,166,168,176,178,180,181,185,186]
14	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	[146,154,155,160,161,165,172,173,177,182,190,191]
15	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	[147,151,156,157,162,167,169,174,179,183,187,192]
16	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[148,152,153,158,159,163,170,171,175,184,188,189]

Table 5: Wyckoff site: `32e`, site symmetry: `.3m`

No.	position	mapping
1	$[x, x, x]$	[1,5,6,20,22,24]
2	$\left[x, \frac{1}{4}-x, \frac{1}{4}-x\right]$	[2,10,11,16,17,21]
3	$\left[\frac{1}{4}-x, x, \frac{1}{4}-x\right]$	[3,7,12,13,18,23]
4	$\left[\frac{1}{4}-x, \frac{1}{4}-x, x\right]$	[4,8,9,14,15,19]
5	$\left[x+\frac{1}{4}, -x, x+\frac{1}{4}\right]$	[25,30,35,39,43,48]
6	$\left[x+\frac{1}{4}, x+\frac{1}{4}, -x\right]$	[26,27,31,40,44,45]
7	$\left[-x, x+\frac{1}{4}, x+\frac{1}{4}\right]$	[28,29,33,38,46,47]
8	$[-x, -x, -x]$	[32,34,36,37,41,42]
9	$\left[x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[49,53,54,68,70,72]
10	$\left[x, \frac{3}{4}-x, \frac{3}{4}-x\right]$	[50,58,59,64,65,69]
11	$\left[\frac{1}{4}-x, x+\frac{1}{2}, \frac{3}{4}-x\right]$	[51,55,60,61,66,71]
12	$\left[\frac{1}{4}-x, \frac{3}{4}-x, x+\frac{1}{2}\right]$	[52,56,57,62,63,67]
13	$\left[x+\frac{1}{4}, \frac{1}{2}-x, x+\frac{3}{4}\right]$	[73,78,83,87,91,96]
14	$\left[x+\frac{1}{4}, x+\frac{3}{4}, \frac{1}{2}-x\right]$	[74,75,79,88,92,93]
15	$\left[-x, x+\frac{3}{4}, x+\frac{3}{4}\right]$	[76,77,81,86,94,95]
16	$\left[-x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[80,82,84,85,89,90]
17	$\left[x+\frac{1}{2}, x, x+\frac{1}{2}\right]$	[97,101,102,116,118,120]
18	$\left[x+\frac{1}{2}, \frac{1}{4}-x, \frac{3}{4}-x\right]$	[98,106,107,112,113,117]
19	$\left[\frac{3}{4}-x, x, \frac{3}{4}-x\right]$	[99,103,108,109,114,119]
20	$\left[\frac{3}{4}-x, \frac{1}{4}-x, x+\frac{1}{2}\right]$	[100,104,105,110,111,115]
21	$\left[x+\frac{3}{4}, -x, x+\frac{3}{4}\right]$	[121,126,131,135,139,144]
22	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{2}-x\right]$	[122,123,127,136,140,141]
23	$\left[\frac{1}{2}-x, x+\frac{1}{4}, x+\frac{3}{4}\right]$	[124,125,129,134,142,143]
24	$\left[\frac{1}{2}-x, -x, \frac{1}{2}-x\right]$	[128,130,132,133,137,138]
25	$\left[x+\frac{1}{2}, x+\frac{1}{2}, x\right]$	[145,149,150,164,166,168]

continued ...

Table 5

No.	position	mapping
26	$\left[x+\frac{1}{2},\ \frac{3}{4}-x,\ \frac{1}{4}-x\right]$	[146,154,155,160,161,165]
27	$\left[\frac{3}{4}-x,\ x+\frac{1}{2},\ \frac{1}{4}-x\right]$	[147,151,156,157,162,167]
28	$\left[\frac{3}{4}-x,\ \frac{3}{4}-x,\ x\right]$	[148,152,153,158,159,163]
29	$\left[x+\frac{3}{4},\ \frac{1}{2}-x,\ x+\frac{1}{4}\right]$	[169,174,179,183,187,192]
30	$\left[x+\frac{3}{4},\ x+\frac{3}{4},\ -x\right]$	[170,171,175,184,188,189]
31	$\left[\frac{1}{2}-x,\ x+\frac{3}{4},\ x+\frac{1}{4}\right]$	[172,173,177,182,190,191]
32	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ -x\right]$	[176,178,180,181,185,186]

Table 6: Wyckoff site: `48f`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[x,\ \frac{1}{8},\ \frac{1}{8}\right]$	[1,2,21,22]
2	$\left[\frac{1}{4}-x,\ \frac{1}{8},\ \frac{1}{8}\right]$	[3,4,13,14]
3	$\left[\frac{1}{8},\ x,\ \frac{1}{8}\right]$	[5,12,18,20]
4	$\left[\frac{1}{8},\ \frac{1}{8},\ x\right]$	[6,9,15,24]
5	$\left[\frac{1}{8},\ \frac{1}{8},\ \frac{1}{4}-x\right]$	[7,11,16,23]
6	$\left[\frac{1}{8},\ \frac{1}{4}-x,\ \frac{1}{8}\right]$	[8,10,17,19]
7	$\left[x+\frac{1}{4},\ \frac{7}{8},\ \frac{3}{8}\right]$	[25,39,74,88]
8	$\left[x+\frac{1}{4},\ \frac{3}{8},\ \frac{7}{8}\right]$	[26,40,73,87]
9	$\left[\frac{3}{8},\ \frac{3}{8},\ -x\right]$	[27,45,180,186]
10	$\left[\frac{7}{8},\ \frac{3}{8},\ x+\frac{1}{4}\right]$	[28,47,179,187]
11	$\left[\frac{7}{8},\ x+\frac{1}{4},\ \frac{3}{8}\right]$	[29,46,127,140]
12	$\left[\frac{3}{8},\ -x,\ \frac{3}{8}\right]$	[30,48,128,137]
13	$\left[\frac{3}{8},\ x+\frac{1}{4},\ \frac{7}{8}\right]$	[31,44,125,142]
14	$\left[\frac{7}{8},\ -x,\ \frac{7}{8}\right]$	[32,41,126,144]
15	$\left[-x,\ \frac{3}{8},\ \frac{3}{8}\right]$	[33,38,82,85]
16	$\left[-x,\ \frac{7}{8},\ \frac{7}{8}\right]$	[34,37,81,86]
17	$\left[\frac{3}{8},\ \frac{7}{8},\ x+\frac{1}{4}\right]$	[35,43,172,191]
18	$\left[\frac{7}{8},\ \frac{7}{8},\ -x\right]$	[36,42,171,189]
19	$\left[x,\ \frac{5}{8},\ \frac{5}{8}\right]$	[49,50,69,70]
20	$\left[\frac{1}{4}-x,\ \frac{5}{8},\ \frac{5}{8}\right]$	[51,52,61,62]
21	$\left[\frac{1}{8},\ x+\frac{1}{2},\ \frac{5}{8}\right]$	[53,60,66,68]
22	$\left[\frac{1}{8},\ \frac{5}{8},\ x+\frac{1}{2}\right]$	[54,57,63,72]
23	$\left[\frac{1}{8},\ \frac{5}{8},\ \frac{3}{4}-x\right]$	[55,59,64,71]
24	$\left[\frac{1}{8},\ \frac{3}{4}-x,\ \frac{5}{8}\right]$	[56,58,65,67]
25	$\left[\frac{3}{8},\ \frac{7}{8},\ \frac{1}{2}-x\right]$	[75,93,132,138]
26	$\left[\frac{7}{8},\ \frac{7}{8},\ x+\frac{3}{4}\right]$	[76,95,131,139]
27	$\left[\frac{7}{8},\ x+\frac{3}{4},\ \frac{7}{8}\right]$	[77,94,175,188]
28	$\left[\frac{3}{8},\ \frac{1}{2}-x,\ \frac{7}{8}\right]$	[78,96,176,185]
29	$\left[\frac{3}{8},\ x+\frac{3}{4},\ \frac{3}{8}\right]$	[79,92,173,190]
30	$\left[\frac{7}{8},\ \frac{1}{2}-x,\ \frac{3}{8}\right]$	[80,89,174,192]
31	$\left[\frac{3}{8},\ \frac{3}{8},\ x+\frac{3}{4}\right]$	[83,91,124,143]
32	$\left[\frac{7}{8},\ \frac{3}{8},\ \frac{1}{2}-x\right]$	[84,90,123,141]
33	$\left[x+\frac{1}{2},\ \frac{1}{8},\ \frac{5}{8}\right]$	[97,98,117,118]
34	$\left[\frac{3}{4}-x,\ \frac{1}{8},\ \frac{5}{8}\right]$	[99,100,109,110]

continued ...

Table 6

No.	position	mapping
35	$\left[\frac{5}{8}, x, \frac{5}{8}\right]$	[101,108,114,116]
36	$\left[\frac{5}{8}, \frac{1}{8}, x+\frac{1}{2}\right]$	[102,105,111,120]
37	$\left[\frac{5}{8}, \frac{1}{8}, \frac{3}{4}-x\right]$	[103,107,112,119]
38	$\left[\frac{5}{8}, \frac{1}{4}-x, \frac{5}{8}\right]$	[104,106,113,115]
39	$\left[x+\frac{3}{4}, \frac{7}{8}, \frac{7}{8}\right]$	[121,135,170,184]
40	$\left[x+\frac{3}{4}, \frac{3}{8}, \frac{3}{8}\right]$	[122,136,169,183]
41	$\left[\frac{1}{2}-x, \frac{3}{8}, \frac{7}{8}\right]$	[129,134,178,181]
42	$\left[\frac{1}{2}-x, \frac{7}{8}, \frac{3}{8}\right]$	[130,133,177,182]
43	$\left[x+\frac{1}{2}, \frac{5}{8}, \frac{1}{8}\right]$	[145,146,165,166]
44	$\left[\frac{3}{4}-x, \frac{5}{8}, \frac{1}{8}\right]$	[147,148,157,158]
45	$\left[\frac{5}{8}, x+\frac{1}{2}, \frac{1}{8}\right]$	[149,156,162,164]
46	$\left[\frac{5}{8}, \frac{5}{8}, x\right]$	[150,153,159,168]
47	$\left[\frac{5}{8}, \frac{5}{8}, \frac{1}{4}-x\right]$	[151,155,160,167]
48	$\left[\frac{5}{8}, \frac{3}{4}-x, \frac{1}{8}\right]$	[152,154,161,163]

Table 7: Wyckoff site: 96g, site symmetry: ..m

No.	position	mapping
1	$[x, x, z]$	[1,20]
2	$\left[x, \frac{1}{4}-x, \frac{1}{4}-z\right]$	[2,17]
3	$\left[\frac{1}{4}-x, x, \frac{1}{4}-z\right]$	[3,18]
4	$\left[\frac{1}{4}-x, \frac{1}{4}-x, z\right]$	[4,19]
5	$[z, x, x]$	[5,24]
6	$[x, z, x]$	[6,22]
7	$\left[\frac{1}{4}-x, z, \frac{1}{4}-x\right]$	[7,13]
8	$\left[\frac{1}{4}-z, \frac{1}{4}-x, x\right]$	[8,15]
9	$\left[\frac{1}{4}-x, \frac{1}{4}-z, x\right]$	[9,14]
10	$\left[z, \frac{1}{4}-x, \frac{1}{4}-x\right]$	[10,16]
11	$\left[x, \frac{1}{4}-z, \frac{1}{4}-x\right]$	[11,21]
12	$\left[\frac{1}{4}-z, x, \frac{1}{4}-x\right]$	[12,23]
13	$\left[x+\frac{1}{4}, -z, x+\frac{1}{4}\right]$	[25,43]
14	$\left[x+\frac{1}{4}, z+\frac{1}{4}, -x\right]$	[26,45]
15	$\left[z+\frac{1}{4}, x+\frac{1}{4}, -x\right]$	[27,44]
16	$\left[-z, x+\frac{1}{4}, x+\frac{1}{4}\right]$	[28,46]
17	$\left[-x, x+\frac{1}{4}, z+\frac{1}{4}\right]$	[29,38]
18	$\left[x+\frac{1}{4}, -x, z+\frac{1}{4}\right]$	[30,39]
19	$\left[x+\frac{1}{4}, x+\frac{1}{4}, -z\right]$	[31,40]
20	$[-x, -x, -z]$	[32,37]
21	$\left[-x, z+\frac{1}{4}, x+\frac{1}{4}\right]$	[33,47]
22	$[-x, -z, -x]$	[34,42]
23	$\left[z+\frac{1}{4}, -x, x+\frac{1}{4}\right]$	[35,48]
24	$[-z, -x, -x]$	[36,41]
25	$\left[x, x+\frac{1}{2}, z+\frac{1}{2}\right]$	[49,68]
26	$\left[x, \frac{3}{4}-x, \frac{3}{4}-z\right]$	[50,65]
27	$\left[\frac{1}{4}-x, x+\frac{1}{2}, \frac{3}{4}-z\right]$	[51,66]

continued ...

Table 7

No.	position	mapping
28	$\left[\frac{1}{4}-x,\ \frac{3}{4}-x,\ z+\frac{1}{2}\right]$	[52,67]
29	$\left[z,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	[53,72]
30	$\left[x,\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$	[54,70]
31	$\left[\frac{1}{4}-x,\ z+\frac{1}{2},\ \frac{3}{4}-x\right]$	[55,61]
32	$\left[\frac{1}{4}-z,\ \frac{3}{4}-x,\ x+\frac{1}{2}\right]$	[56,63]
33	$\left[\frac{1}{4}-x,\ \frac{3}{4}-z,\ x+\frac{1}{2}\right]$	[57,62]
34	$\left[z,\ \frac{3}{4}-x,\ \frac{3}{4}-x\right]$	[58,64]
35	$\left[x,\ \frac{3}{4}-z,\ \frac{3}{4}-x\right]$	[59,69]
36	$\left[\frac{1}{4}-z,\ x+\frac{1}{2},\ \frac{3}{4}-x\right]$	[60,71]
37	$\left[x+\frac{1}{4},\ \frac{1}{2}-z,\ x+\frac{3}{4}\right]$	[73,91]
38	$\left[x+\frac{1}{4},\ z+\frac{3}{4},\ \frac{1}{2}-x\right]$	[74,93]
39	$\left[z+\frac{1}{4},\ x+\frac{3}{4},\ \frac{1}{2}-x\right]$	[75,92]
40	$\left[-z,\ x+\frac{3}{4},\ x+\frac{3}{4}\right]$	[76,94]
41	$\left[-x,\ x+\frac{3}{4},\ z+\frac{3}{4}\right]$	[77,86]
42	$\left[x+\frac{1}{4},\ \frac{1}{2}-x,\ z+\frac{3}{4}\right]$	[78,87]
43	$\left[x+\frac{1}{4},\ x+\frac{3}{4},\ \frac{1}{2}-z\right]$	[79,88]
44	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[80,85]
45	$\left[-x,\ z+\frac{3}{4},\ x+\frac{3}{4}\right]$	[81,95]
46	$\left[-x,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	[82,90]
47	$\left[z+\frac{1}{4},\ \frac{1}{2}-x,\ x+\frac{3}{4}\right]$	[83,96]
48	$\left[-z,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	[84,89]
49	$\left[x+\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	[97,116]
50	$\left[x+\frac{1}{2},\ \frac{1}{4}-x,\ \frac{3}{4}-z\right]$	[98,113]
51	$\left[\frac{3}{4}-x,\ x,\ \frac{3}{4}-z\right]$	[99,114]
52	$\left[\frac{3}{4}-x,\ \frac{1}{4}-x,\ z+\frac{1}{2}\right]$	[100,115]
53	$\left[z+\frac{1}{2},\ x,\ x+\frac{1}{2}\right]$	[101,120]
54	$\left[x+\frac{1}{2},\ z,\ x+\frac{1}{2}\right]$	[102,118]
55	$\left[\frac{3}{4}-x,\ z,\ \frac{3}{4}-x\right]$	[103,109]
56	$\left[\frac{3}{4}-z,\ \frac{1}{4}-x,\ x+\frac{1}{2}\right]$	[104,111]
57	$\left[\frac{3}{4}-x,\ \frac{1}{4}-z,\ x+\frac{1}{2}\right]$	[105,110]
58	$\left[z+\frac{1}{2},\ \frac{1}{4}-x,\ \frac{3}{4}-x\right]$	[106,112]
59	$\left[x+\frac{1}{2},\ \frac{1}{4}-z,\ \frac{3}{4}-x\right]$	[107,117]
60	$\left[\frac{3}{4}-z,\ x,\ \frac{3}{4}-x\right]$	[108,119]
61	$\left[x+\frac{3}{4},\ -z,\ x+\frac{3}{4}\right]$	[121,139]
62	$\left[x+\frac{3}{4},\ z+\frac{1}{4},\ \frac{1}{2}-x\right]$	[122,141]
63	$\left[z+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{2}-x\right]$	[123,140]
64	$\left[\frac{1}{2}-z,\ x+\frac{1}{4},\ x+\frac{3}{4}\right]$	[124,142]
65	$\left[\frac{1}{2}-x,\ x+\frac{1}{4},\ z+\frac{3}{4}\right]$	[125,134]
66	$\left[x+\frac{3}{4},\ -x,\ z+\frac{3}{4}\right]$	[126,135]
67	$\left[x+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{2}-z\right]$	[127,136]
68	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}-z\right]$	[128,133]
69	$\left[\frac{1}{2}-x,\ z+\frac{1}{4},\ x+\frac{3}{4}\right]$	[129,143]
70	$\left[\frac{1}{2}-x,\ -z,\ \frac{1}{2}-x\right]$	[130,138]
71	$\left[z+\frac{3}{4},\ -x,\ x+\frac{3}{4}\right]$	[131,144]
72	$\left[\frac{1}{2}-z,\ -x,\ \frac{1}{2}-x\right]$	[132,137]
73	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	[145,164]
74	$\left[x+\frac{1}{2},\ \frac{3}{4}-x,\ \frac{1}{4}-z\right]$	[146,161]

continued ...

Table 7

No.	position	mapping
75	$\left[\frac{3}{4}-x,\ x+\frac{1}{2},\ \frac{1}{4}-z\right]$	[147,162]
76	$\left[\frac{3}{4}-x,\ \frac{3}{4}-x,\ z\right]$	[148,163]
77	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ x\right]$	[149,168]
78	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ x\right]$	[150,166]
79	$\left[\frac{3}{4}-x,\ z+\frac{1}{2},\ \frac{1}{4}-x\right]$	[151,157]
80	$\left[\frac{3}{4}-z,\ \frac{3}{4}-x,\ x\right]$	[152,159]
81	$\left[\frac{3}{4}-x,\ \frac{3}{4}-z,\ x\right]$	[153,158]
82	$\left[z+\frac{1}{2},\ \frac{3}{4}-x,\ \frac{1}{4}-x\right]$	[154,160]
83	$\left[x+\frac{1}{2},\ \frac{3}{4}-z,\ \frac{1}{4}-x\right]$	[155,165]
84	$\left[\frac{3}{4}-z,\ x+\frac{1}{2},\ \frac{1}{4}-x\right]$	[156,167]
85	$\left[x+\frac{3}{4},\ \frac{1}{2}-z,\ x+\frac{1}{4}\right]$	[169,187]
86	$\left[x+\frac{3}{4},\ z+\frac{3}{4},\ -x\right]$	[170,189]
87	$\left[z+\frac{3}{4},\ x+\frac{3}{4},\ -x\right]$	[171,188]
88	$\left[\frac{1}{2}-z,\ x+\frac{3}{4},\ x+\frac{1}{4}\right]$	[172,190]
89	$\left[\frac{1}{2}-x,\ x+\frac{3}{4},\ z+\frac{1}{4}\right]$	[173,182]
90	$\left[x+\frac{3}{4},\ \frac{1}{2}-x,\ z+\frac{1}{4}\right]$	[174,183]
91	$\left[x+\frac{3}{4},\ x+\frac{3}{4},\ -z\right]$	[175,184]
92	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ -z\right]$	[176,181]
93	$\left[\frac{1}{2}-x,\ z+\frac{3}{4},\ x+\frac{1}{4}\right]$	[177,191]
94	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ -x\right]$	[178,186]
95	$\left[z+\frac{3}{4},\ \frac{1}{2}-x,\ x+\frac{1}{4}\right]$	[179,192]
96	$\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ -x\right]$	[180,185]

Table 8: Wyckoff site: 96h, site symmetry: ..2'

No.	position	mapping
1	$[0,\ y,\ -y]$	[1,34]
2	$\left[0,\ \frac{1}{4}-y,\ y+\frac{1}{4}\right]$	[2,33]
3	$\left[\frac{1}{4},\ y,\ y+\frac{1}{4}\right]$	[3,25]
4	$\left[\frac{1}{4},\ \frac{1}{4}-y,\ -y\right]$	[4,26]
5	$[-y,\ 0,\ y]$	[5,32]
6	$[y,\ -y,\ 0]$	[6,36]
7	$\left[\frac{1}{4}-y,\ -y,\ \frac{1}{4}\right]$	[7,35]
8	$\left[y+\frac{1}{4},\ \frac{1}{4},\ y\right]$	[8,31]
9	$\left[\frac{1}{4}-y,\ y+\frac{1}{4},\ 0\right]$	[9,27]
10	$\left[-y,\ \frac{1}{4},\ \frac{1}{4}-y\right]$	[10,29]
11	$\left[y,\ y+\frac{1}{4},\ \frac{1}{4}\right]$	[11,28]
12	$\left[y+\frac{1}{4},\ 0,\ \frac{1}{4}-y\right]$	[12,30]
13	$\left[\frac{1}{4},\ -y,\ \frac{1}{4}-y\right]$	[13,39]
14	$\left[\frac{1}{4},\ y+\frac{1}{4},\ y\right]$	[14,40]
15	$\left[y+\frac{1}{4},\ \frac{1}{4}-y,\ 0\right]$	[15,45]
16	$\left[-y,\ \frac{1}{4}-y,\ \frac{1}{4}\right]$	[16,47]
17	$\left[y,\ \frac{1}{4},\ y+\frac{1}{4}\right]$	[17,46]
18	$\left[\frac{1}{4}-y,\ 0,\ y+\frac{1}{4}\right]$	[18,48]
19	$\left[\frac{1}{4}-y,\ \frac{1}{4},\ -y\right]$	[19,44]

continued ...

Table 8

No.	position	mapping
20	$[y,\ 0,\ -y]$	[20,41]
21	$\left[0,\ y+\frac{1}{4},\ \frac{1}{4}-y\right]$	[21,38]
22	$[0,\ -y,\ y]$	[22,37]
23	$\left[y+\frac{1}{4},\ y,\ \frac{1}{4}\right]$	[23,43]
24	$[-y,\ y,\ 0]$	[24,42]
25	$\left[0,\ y+\frac{1}{2},\ \frac{1}{2}-y\right]$	[49,82]
26	$\left[0,\ \frac{3}{4}-y,\ y+\frac{3}{4}\right]$	[50,81]
27	$\left[\frac{1}{4},\ y+\frac{1}{2},\ y+\frac{3}{4}\right]$	[51,73]
28	$\left[\frac{1}{4},\ \frac{3}{4}-y,\ \frac{1}{2}-y\right]$	[52,74]
29	$\left[-y,\ \frac{1}{2},\ y+\frac{1}{2}\right]$	[53,80]
30	$\left[y,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[54,84]
31	$\left[\frac{1}{4}-y,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[55,83]
32	$\left[y+\frac{1}{4},\ \frac{3}{4},\ y+\frac{1}{2}\right]$	[56,79]
33	$\left[\frac{1}{4}-y,\ y+\frac{3}{4},\ \frac{1}{2}\right]$	[57,75]
34	$\left[-y,\ \frac{3}{4},\ \frac{3}{4}-y\right]$	[58,77]
35	$\left[y,\ y+\frac{3}{4},\ \frac{3}{4}\right]$	[59,76]
36	$\left[y+\frac{1}{4},\ \frac{1}{2},\ \frac{3}{4}-y\right]$	[60,78]
37	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{3}{4}-y\right]$	[61,87]
38	$\left[\frac{1}{4},\ y+\frac{3}{4},\ y+\frac{1}{2}\right]$	[62,88]
39	$\left[y+\frac{1}{4},\ \frac{3}{4}-y,\ \frac{1}{2}\right]$	[63,93]
40	$\left[-y,\ \frac{3}{4}-y,\ \frac{3}{4}\right]$	[64,95]
41	$\left[y,\ \frac{3}{4},\ y+\frac{3}{4}\right]$	[65,94]
42	$\left[\frac{1}{4}-y,\ \frac{1}{2},\ y+\frac{3}{4}\right]$	[66,96]
43	$\left[\frac{1}{4}-y,\ \frac{3}{4},\ \frac{1}{2}-y\right]$	[67,92]
44	$\left[y,\ \frac{1}{2},\ \frac{1}{2}-y\right]$	[68,89]
45	$\left[0,\ y+\frac{3}{4},\ \frac{3}{4}-y\right]$	[69,86]
46	$\left[0,\ \frac{1}{2}-y,\ y+\frac{1}{2}\right]$	[70,85]
47	$\left[y+\frac{1}{4},\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[71,91]
48	$\left[-y,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[72,90]
49	$\left[\frac{1}{2},\ y,\ \frac{1}{2}-y\right]$	[97,130]
50	$\left[\frac{1}{2},\ \frac{1}{4}-y,\ y+\frac{3}{4}\right]$	[98,129]
51	$\left[\frac{3}{4},\ y,\ y+\frac{3}{4}\right]$	[99,121]
52	$\left[\frac{3}{4},\ \frac{1}{4}-y,\ \frac{1}{2}-y\right]$	[100,122]
53	$\left[\frac{1}{2}-y,\ 0,\ y+\frac{1}{2}\right]$	[101,128]
54	$\left[y+\frac{1}{2},\ -y,\ \frac{1}{2}\right]$	[102,132]
55	$\left[\frac{3}{4}-y,\ -y,\ \frac{3}{4}\right]$	[103,131]
56	$\left[y+\frac{3}{4},\ \frac{1}{4},\ y+\frac{1}{2}\right]$	[104,127]
57	$\left[\frac{3}{4}-y,\ y+\frac{1}{4},\ \frac{1}{2}\right]$	[105,123]
58	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ \frac{3}{4}-y\right]$	[106,125]
59	$\left[y+\frac{1}{2},\ y+\frac{1}{4},\ \frac{3}{4}\right]$	[107,124]
60	$\left[y+\frac{3}{4},\ 0,\ \frac{3}{4}-y\right]$	[108,126]
61	$\left[\frac{3}{4},\ -y,\ \frac{3}{4}-y\right]$	[109,135]
62	$\left[\frac{3}{4},\ y+\frac{1}{4},\ y+\frac{1}{2}\right]$	[110,136]
63	$\left[y+\frac{3}{4},\ \frac{1}{4}-y,\ \frac{1}{2}\right]$	[111,141]
64	$\left[\frac{1}{2}-y,\ \frac{1}{4}-y,\ \frac{3}{4}\right]$	[112,143]
65	$\left[y+\frac{1}{2},\ \frac{1}{4},\ y+\frac{3}{4}\right]$	[113,142]
66	$\left[\frac{3}{4}-y,\ 0,\ y+\frac{3}{4}\right]$	[114,144]

continued ...

Table 8

No.	position	mapping
67	$\left[\frac{3}{4}-y,\ \frac{1}{4},\ \frac{1}{2}-y\right]$	[115,140]
68	$\left[y+\frac{1}{2},\ 0,\ \frac{1}{2}-y\right]$	[116,137]
69	$\left[\frac{1}{2},\ y+\frac{1}{4},\ \frac{3}{4}-y\right]$	[117,134]
70	$\left[\frac{1}{2},\ -y,\ y+\frac{1}{2}\right]$	[118,133]
71	$\left[y+\frac{3}{4},\ y,\ \frac{3}{4}\right]$	[119,139]
72	$\left[\frac{1}{2}-y,\ y,\ \frac{1}{2}\right]$	[120,138]
73	$\left[\frac{1}{2},\ y+\frac{1}{2},\ -y\right]$	[145,178]
74	$\left[\frac{1}{2},\ \frac{3}{4}-y,\ y+\frac{1}{4}\right]$	[146,177]
75	$\left[\frac{3}{4},\ y+\frac{1}{2},\ y+\frac{1}{4}\right]$	[147,169]
76	$\left[\frac{3}{4},\ \frac{3}{4}-y,\ -y\right]$	[148,170]
77	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ y\right]$	[149,176]
78	$\left[y+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	[150,180]
79	$\left[\frac{3}{4}-y,\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[151,179]
80	$\left[y+\frac{3}{4},\ \frac{3}{4},\ y\right]$	[152,175]
81	$\left[\frac{3}{4}-y,\ y+\frac{3}{4},\ 0\right]$	[153,171]
82	$\left[\frac{1}{2}-y,\ \frac{3}{4},\ \frac{1}{4}-y\right]$	[154,173]
83	$\left[y+\frac{1}{2},\ y+\frac{3}{4},\ \frac{1}{4}\right]$	[155,172]
84	$\left[y+\frac{3}{4},\ \frac{1}{2},\ \frac{1}{4}-y\right]$	[156,174]
85	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ \frac{1}{4}-y\right]$	[157,183]
86	$\left[\frac{3}{4},\ y+\frac{3}{4},\ y\right]$	[158,184]
87	$\left[y+\frac{3}{4},\ \frac{3}{4}-y,\ 0\right]$	[159,189]
88	$\left[\frac{1}{2}-y,\ \frac{3}{4}-y,\ \frac{1}{4}\right]$	[160,191]
89	$\left[y+\frac{1}{2},\ \frac{3}{4},\ y+\frac{1}{4}\right]$	[161,190]
90	$\left[\frac{3}{4}-y,\ \frac{1}{2},\ y+\frac{1}{4}\right]$	[162,192]
91	$\left[\frac{3}{4}-y,\ \frac{3}{4},\ -y\right]$	[163,188]
92	$\left[y+\frac{1}{2},\ \frac{1}{2},\ -y\right]$	[164,185]
93	$\left[\frac{1}{2},\ y+\frac{3}{4},\ \frac{1}{4}-y\right]$	[165,182]
94	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ y\right]$	[166,181]
95	$\left[y+\frac{3}{4},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[167,187]
96	$\left[\frac{1}{2}-y,\ y+\frac{1}{2},\ 0\right]$	[168,186]

Table 9: Wyckoff site: 192i, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[x,\ \frac{1}{4}-y,\ \frac{1}{4}-z\right]$	[2]
3	$\left[\frac{1}{4}-x,\ y,\ \frac{1}{4}-z\right]$	[3]
4	$\left[\frac{1}{4}-x,\ \frac{1}{4}-y,\ z\right]$	[4]
5	$[z,\ x,\ y]$	[5]
6	$[y,\ z,\ x]$	[6]
7	$\left[\frac{1}{4}-y,\ z,\ \frac{1}{4}-x\right]$	[7]
8	$\left[\frac{1}{4}-z,\ \frac{1}{4}-x,\ y\right]$	[8]
9	$\left[\frac{1}{4}-y,\ \frac{1}{4}-z,\ x\right]$	[9]
10	$\left[z,\ \frac{1}{4}-x,\ \frac{1}{4}-y\right]$	[10]
11	$\left[y,\ \frac{1}{4}-z,\ \frac{1}{4}-x\right]$	[11]

continued ...

Table 9

No.	position	mapping
12	$\left[\frac{1}{4}-z,\, x,\, \frac{1}{4}-y\right]$	[12]
13	$\left[\frac{1}{4}-x,\, z,\, \frac{1}{4}-y\right]$	[13]
14	$\left[\frac{1}{4}-x,\, \frac{1}{4}-z,\, y\right]$	[14]
15	$\left[\frac{1}{4}-z,\, \frac{1}{4}-y,\, x\right]$	[15]
16	$\left[z,\, \frac{1}{4}-y,\, \frac{1}{4}-x\right]$	[16]
17	$\left[y,\, \frac{1}{4}-x,\, \frac{1}{4}-z\right]$	[17]
18	$\left[\frac{1}{4}-y,\, x,\, \frac{1}{4}-z\right]$	[18]
19	$\left[\frac{1}{4}-y,\, \frac{1}{4}-x,\, z\right]$	[19]
20	$[y,\, x,\, z]$	[20]
21	$\left[x,\, \frac{1}{4}-z,\, \frac{1}{4}-y\right]$	[21]
22	$[x,\, z,\, y]$	[22]
23	$\left[\frac{1}{4}-z,\, y,\, \frac{1}{4}-x\right]$	[23]
24	$[z,\, y,\, x]$	[24]
25	$\left[x+\frac{1}{4},\, -z,\, y+\frac{1}{4}\right]$	[25]
26	$\left[x+\frac{1}{4},\, z+\frac{1}{4},\, -y\right]$	[26]
27	$\left[z+\frac{1}{4},\, y+\frac{1}{4},\, -x\right]$	[27]
28	$\left[-z,\, y+\frac{1}{4},\, x+\frac{1}{4}\right]$	[28]
29	$\left[-y,\, x+\frac{1}{4},\, z+\frac{1}{4}\right]$	[29]
30	$\left[y+\frac{1}{4},\, -x,\, z+\frac{1}{4}\right]$	[30]
31	$\left[y+\frac{1}{4},\, x+\frac{1}{4},\, -z\right]$	[31]
32	$[-y,\, -x,\, -z]$	[32]
33	$\left[-x,\, z+\frac{1}{4},\, y+\frac{1}{4}\right]$	[33]
34	$[-x,\, -z,\, -y]$	[34]
35	$\left[z+\frac{1}{4},\, -y,\, x+\frac{1}{4}\right]$	[35]
36	$[-z,\, -y,\, -x]$	[36]
37	$[-x,\, -y,\, -z]$	[37]
38	$\left[-x,\, y+\frac{1}{4},\, z+\frac{1}{4}\right]$	[38]
39	$\left[x+\frac{1}{4},\, -y,\, z+\frac{1}{4}\right]$	[39]
40	$\left[x+\frac{1}{4},\, y+\frac{1}{4},\, -z\right]$	[40]
41	$[-z,\, -x,\, -y]$	[41]
42	$[-y,\, -z,\, -x]$	[42]
43	$\left[y+\frac{1}{4},\, -z,\, x+\frac{1}{4}\right]$	[43]
44	$\left[z+\frac{1}{4},\, x+\frac{1}{4},\, -y\right]$	[44]
45	$\left[y+\frac{1}{4},\, z+\frac{1}{4},\, -x\right]$	[45]
46	$\left[-z,\, x+\frac{1}{4},\, y+\frac{1}{4}\right]$	[46]
47	$\left[-y,\, z+\frac{1}{4},\, x+\frac{1}{4}\right]$	[47]
48	$\left[z+\frac{1}{4},\, -x,\, y+\frac{1}{4}\right]$	[48]
49	$\left[x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[49]
50	$\left[x,\, \frac{3}{4}-y,\, \frac{3}{4}-z\right]$	[50]
51	$\left[\frac{1}{4}-x,\, y+\frac{1}{2},\, \frac{3}{4}-z\right]$	[51]
52	$\left[\frac{1}{4}-x,\, \frac{3}{4}-y,\, z+\frac{1}{2}\right]$	[52]
53	$\left[z,\, x+\frac{1}{2},\, y+\frac{1}{2}\right]$	[53]
54	$\left[y,\, z+\frac{1}{2},\, x+\frac{1}{2}\right]$	[54]
55	$\left[\frac{1}{4}-y,\, z+\frac{1}{2},\, \frac{3}{4}-x\right]$	[55]
56	$\left[\frac{1}{4}-z,\, \frac{3}{4}-x,\, y+\frac{1}{2}\right]$	[56]
57	$\left[\frac{1}{4}-y,\, \frac{3}{4}-z,\, x+\frac{1}{2}\right]$	[57]
58	$\left[z,\, \frac{3}{4}-x,\, \frac{3}{4}-y\right]$	[58]

continued ...

Table 9

No.	position	mapping
59	$\left[y,\ \frac{3}{4}-z,\ \frac{3}{4}-x\right]$	[59]
60	$\left[\frac{1}{4}-z,\ x+\frac{1}{2},\ \frac{3}{4}-y\right]$	[60]
61	$\left[\frac{1}{4}-x,\ z+\frac{1}{2},\ \frac{3}{4}-y\right]$	[61]
62	$\left[\frac{1}{4}-x,\ \frac{3}{4}-z,\ y+\frac{1}{2}\right]$	[62]
63	$\left[\frac{1}{4}-z,\ \frac{3}{4}-y,\ x+\frac{1}{2}\right]$	[63]
64	$\left[z,\ \frac{3}{4}-y,\ \frac{3}{4}-x\right]$	[64]
65	$\left[y,\ \frac{3}{4}-x,\ \frac{3}{4}-z\right]$	[65]
66	$\left[\frac{1}{4}-y,\ x+\frac{1}{2},\ \frac{3}{4}-z\right]$	[66]
67	$\left[\frac{1}{4}-y,\ \frac{3}{4}-x,\ z+\frac{1}{2}\right]$	[67]
68	$\left[y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[68]
69	$\left[x,\ \frac{3}{4}-z,\ \frac{3}{4}-y\right]$	[69]
70	$\left[x,\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$	[70]
71	$\left[\frac{1}{4}-z,\ y+\frac{1}{2},\ \frac{3}{4}-x\right]$	[71]
72	$\left[z,\ y+\frac{1}{2},\ x+\frac{1}{2}\right]$	[72]
73	$\left[x+\frac{1}{4},\ \frac{1}{2}-z,\ y+\frac{3}{4}\right]$	[73]
74	$\left[x+\frac{1}{4},\ z+\frac{3}{4},\ \frac{1}{2}-y\right]$	[74]
75	$\left[z+\frac{1}{4},\ y+\frac{3}{4},\ \frac{1}{2}-x\right]$	[75]
76	$\left[-z,\ y+\frac{3}{4},\ x+\frac{3}{4}\right]$	[76]
77	$\left[-y,\ x+\frac{3}{4},\ z+\frac{3}{4}\right]$	[77]
78	$\left[y+\frac{1}{4},\ \frac{1}{2}-x,\ z+\frac{3}{4}\right]$	[78]
79	$\left[y+\frac{1}{4},\ x+\frac{3}{4},\ \frac{1}{2}-z\right]$	[79]
80	$\left[-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[80]
81	$\left[-x,\ z+\frac{3}{4},\ y+\frac{3}{4}\right]$	[81]
82	$\left[-x,\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$	[82]
83	$\left[z+\frac{1}{4},\ \frac{1}{2}-y,\ x+\frac{3}{4}\right]$	[83]
84	$\left[-z,\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$	[84]
85	$\left[-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[85]
86	$\left[-x,\ y+\frac{3}{4},\ z+\frac{3}{4}\right]$	[86]
87	$\left[x+\frac{1}{4},\ \frac{1}{2}-y,\ z+\frac{3}{4}\right]$	[87]
88	$\left[x+\frac{1}{4},\ y+\frac{3}{4},\ \frac{1}{2}-z\right]$	[88]
89	$\left[-z,\ \frac{1}{2}-x,\ \frac{1}{2}-y\right]$	[89]
90	$\left[-y,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	[90]
91	$\left[y+\frac{1}{4},\ \frac{1}{2}-z,\ x+\frac{3}{4}\right]$	[91]
92	$\left[z+\frac{1}{4},\ x+\frac{3}{4},\ \frac{1}{2}-y\right]$	[92]
93	$\left[y+\frac{1}{4},\ z+\frac{3}{4},\ \frac{1}{2}-x\right]$	[93]
94	$\left[-z,\ x+\frac{3}{4},\ y+\frac{3}{4}\right]$	[94]
95	$\left[-y,\ z+\frac{3}{4},\ x+\frac{3}{4}\right]$	[95]
96	$\left[z+\frac{1}{4},\ \frac{1}{2}-x,\ y+\frac{3}{4}\right]$	[96]
97	$\left[x+\frac{1}{2},\ y,\ z+\frac{1}{2}\right]$	[97]
98	$\left[x+\frac{1}{2},\ \frac{1}{4}-y,\ \frac{3}{4}-z\right]$	[98]
99	$\left[\frac{3}{4}-x,\ y,\ \frac{3}{4}-z\right]$	[99]
100	$\left[\frac{3}{4}-x,\ \frac{1}{4}-y,\ z+\frac{1}{2}\right]$	[100]
101	$\left[z+\frac{1}{2},\ x,\ y+\frac{1}{2}\right]$	[101]
102	$\left[y+\frac{1}{2},\ z,\ x+\frac{1}{2}\right]$	[102]
103	$\left[\frac{3}{4}-y,\ z,\ \frac{3}{4}-x\right]$	[103]
104	$\left[\frac{3}{4}-z,\ \frac{1}{4}-x,\ y+\frac{1}{2}\right]$	[104]
105	$\left[\frac{3}{4}-y,\ \frac{1}{4}-z,\ x+\frac{1}{2}\right]$	[105]

continued ...

Table 9

No.	position	mapping
106	$\left[z+\frac{1}{2},\ \frac{1}{4}-x,\ \frac{3}{4}-y\right]$	[106]
107	$\left[y+\frac{1}{2},\ \frac{1}{4}-z,\ \frac{3}{4}-x\right]$	[107]
108	$\left[\frac{3}{4}-z,\ x,\ \frac{3}{4}-y\right]$	[108]
109	$\left[\frac{3}{4}-x,\ z,\ \frac{3}{4}-y\right]$	[109]
110	$\left[\frac{3}{4}-x,\ \frac{1}{4}-z,\ y+\frac{1}{2}\right]$	[110]
111	$\left[\frac{3}{4}-z,\ \frac{1}{4}-y,\ x+\frac{1}{2}\right]$	[111]
112	$\left[z+\frac{1}{2},\ \frac{1}{4}-y,\ \frac{3}{4}-x\right]$	[112]
113	$\left[y+\frac{1}{2},\ \frac{1}{4}-x,\ \frac{3}{4}-z\right]$	[113]
114	$\left[\frac{3}{4}-y,\ x,\ \frac{3}{4}-z\right]$	[114]
115	$\left[\frac{3}{4}-y,\ \frac{1}{4}-x,\ z+\frac{1}{2}\right]$	[115]
116	$\left[y+\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	[116]
117	$\left[x+\frac{1}{2},\ \frac{1}{4}-z,\ \frac{3}{4}-y\right]$	[117]
118	$\left[x+\frac{1}{2},\ z,\ y+\frac{1}{2}\right]$	[118]
119	$\left[\frac{3}{4}-z,\ y,\ \frac{3}{4}-x\right]$	[119]
120	$\left[z+\frac{1}{2},\ y,\ x+\frac{1}{2}\right]$	[120]
121	$\left[x+\frac{3}{4},\ -z,\ y+\frac{3}{4}\right]$	[121]
122	$\left[x+\frac{3}{4},\ z+\frac{1}{4},\ \frac{1}{2}-y\right]$	[122]
123	$\left[z+\frac{3}{4},\ y+\frac{1}{4},\ \frac{1}{2}-x\right]$	[123]
124	$\left[\frac{1}{2}-z,\ y+\frac{1}{4},\ x+\frac{3}{4}\right]$	[124]
125	$\left[\frac{1}{2}-y,\ x+\frac{1}{4},\ z+\frac{3}{4}\right]$	[125]
126	$\left[y+\frac{3}{4},\ -x,\ z+\frac{3}{4}\right]$	[126]
127	$\left[y+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{2}-z\right]$	[127]
128	$\left[\frac{1}{2}-y,\ -x,\ \frac{1}{2}-z\right]$	[128]
129	$\left[\frac{1}{2}-x,\ z+\frac{1}{4},\ y+\frac{3}{4}\right]$	[129]
130	$\left[\frac{1}{2}-x,\ -z,\ \frac{1}{2}-y\right]$	[130]
131	$\left[z+\frac{3}{4},\ -y,\ x+\frac{3}{4}\right]$	[131]
132	$\left[\frac{1}{2}-z,\ -y,\ \frac{1}{2}-x\right]$	[132]
133	$\left[\frac{1}{2}-x,\ -y,\ \frac{1}{2}-z\right]$	[133]
134	$\left[\frac{1}{2}-x,\ y+\frac{1}{4},\ z+\frac{3}{4}\right]$	[134]
135	$\left[x+\frac{3}{4},\ -y,\ z+\frac{3}{4}\right]$	[135]
136	$\left[x+\frac{3}{4},\ y+\frac{1}{4},\ \frac{1}{2}-z\right]$	[136]
137	$\left[\frac{1}{2}-z,\ -x,\ \frac{1}{2}-y\right]$	[137]
138	$\left[\frac{1}{2}-y,\ -z,\ \frac{1}{2}-x\right]$	[138]
139	$\left[y+\frac{3}{4},\ -z,\ x+\frac{3}{4}\right]$	[139]
140	$\left[z+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{2}-y\right]$	[140]
141	$\left[y+\frac{3}{4},\ z+\frac{1}{4},\ \frac{1}{2}-x\right]$	[141]
142	$\left[\frac{1}{2}-z,\ x+\frac{1}{4},\ y+\frac{3}{4}\right]$	[142]
143	$\left[\frac{1}{2}-y,\ z+\frac{1}{4},\ x+\frac{3}{4}\right]$	[143]
144	$\left[z+\frac{3}{4},\ -x,\ y+\frac{3}{4}\right]$	[144]
145	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[145]
146	$\left[x+\frac{1}{2},\ \frac{3}{4}-y,\ \frac{1}{4}-z\right]$	[146]
147	$\left[\frac{3}{4}-x,\ y+\frac{1}{2},\ \frac{1}{4}-z\right]$	[147]
148	$\left[\frac{3}{4}-x,\ \frac{3}{4}-y,\ z\right]$	[148]
149	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ y\right]$	[149]
150	$\left[y+\frac{1}{2},\ z+\frac{1}{2},\ x\right]$	[150]
151	$\left[\frac{3}{4}-y,\ z+\frac{1}{2},\ \frac{1}{4}-x\right]$	[151]
152	$\left[\frac{3}{4}-z,\ \frac{3}{4}-x,\ y\right]$	[152]

continued ...

Table 9

No.	position	mapping
153	$\left[\frac{3}{4}-y,\ \frac{3}{4}-z,\ x\right]$	[153]
154	$\left[z+\frac{1}{2},\ \frac{3}{4}-x,\ \frac{1}{4}-y\right]$	[154]
155	$\left[y+\frac{1}{2},\ \frac{3}{4}-z,\ \frac{1}{4}-x\right]$	[155]
156	$\left[\frac{3}{4}-z,\ x+\frac{1}{2},\ \frac{1}{4}-y\right]$	[156]
157	$\left[\frac{3}{4}-x,\ z+\frac{1}{2},\ \frac{1}{4}-y\right]$	[157]
158	$\left[\frac{3}{4}-x,\ \frac{3}{4}-z,\ y\right]$	[158]
159	$\left[\frac{3}{4}-z,\ \frac{3}{4}-y,\ x\right]$	[159]
160	$\left[z+\frac{1}{2},\ \frac{3}{4}-y,\ \frac{1}{4}-x\right]$	[160]
161	$\left[y+\frac{1}{2},\ \frac{3}{4}-x,\ \frac{1}{4}-z\right]$	[161]
162	$\left[\frac{3}{4}-y,\ x+\frac{1}{2},\ \frac{1}{4}-z\right]$	[162]
163	$\left[\frac{3}{4}-y,\ \frac{3}{4}-x,\ z\right]$	[163]
164	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	[164]
165	$\left[x+\frac{1}{2},\ \frac{3}{4}-z,\ \frac{1}{4}-y\right]$	[165]
166	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ y\right]$	[166]
167	$\left[\frac{3}{4}-z,\ y+\frac{1}{2},\ \frac{1}{4}-x\right]$	[167]
168	$\left[z+\frac{1}{2},\ y+\frac{1}{2},\ x\right]$	[168]
169	$\left[x+\frac{3}{4},\ \frac{1}{2}-z,\ y+\frac{1}{4}\right]$	[169]
170	$\left[x+\frac{3}{4},\ z+\frac{3}{4},\ -y\right]$	[170]
171	$\left[z+\frac{3}{4},\ y+\frac{3}{4},\ -x\right]$	[171]
172	$\left[\frac{1}{2}-z,\ y+\frac{3}{4},\ x+\frac{1}{4}\right]$	[172]
173	$\left[\frac{1}{2}-y,\ x+\frac{3}{4},\ z+\frac{1}{4}\right]$	[173]
174	$\left[y+\frac{3}{4},\ \frac{1}{2}-x,\ z+\frac{1}{4}\right]$	[174]
175	$\left[y+\frac{3}{4},\ x+\frac{3}{4},\ -z\right]$	[175]
176	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ -z\right]$	[176]
177	$\left[\frac{1}{2}-x,\ z+\frac{3}{4},\ y+\frac{1}{4}\right]$	[177]
178	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ -y\right]$	[178]
179	$\left[z+\frac{3}{4},\ \frac{1}{2}-y,\ x+\frac{1}{4}\right]$	[179]
180	$\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ -x\right]$	[180]
181	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ -z\right]$	[181]
182	$\left[\frac{1}{2}-x,\ y+\frac{3}{4},\ z+\frac{1}{4}\right]$	[182]
183	$\left[x+\frac{3}{4},\ \frac{1}{2}-y,\ z+\frac{1}{4}\right]$	[183]
184	$\left[x+\frac{3}{4},\ y+\frac{3}{4},\ -z\right]$	[184]
185	$\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ -y\right]$	[185]
186	$\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ -x\right]$	[186]
187	$\left[y+\frac{3}{4},\ \frac{1}{2}-z,\ x+\frac{1}{4}\right]$	[187]
188	$\left[z+\frac{3}{4},\ x+\frac{3}{4},\ -y\right]$	[188]
189	$\left[y+\frac{3}{4},\ z+\frac{3}{4},\ -x\right]$	[189]
190	$\left[\frac{1}{2}-z,\ x+\frac{3}{4},\ y+\frac{1}{4}\right]$	[190]
191	$\left[\frac{1}{2}-y,\ z+\frac{3}{4},\ x+\frac{1}{4}\right]$	[191]
192	$\left[z+\frac{3}{4},\ \frac{1}{2}-x,\ y+\frac{1}{4}\right]$	[192]